# Fine-Tuning Large Models for 2D Pose Estimation (2024–2025)

Recent top-tier vision conferences (CVPR, ICCV, ECCV 2024–2025) have showcased **foundation models** for human pose estimation and new fine-tuning methods. These approaches build on large pre-trained backbones (e.g. Vision Transformers) and adapt them to 2D keypoint detection with careful fine-tuning strategies. Below, we summarize representative works (e.g. **ViTPose++, Sapiens** from Meta) and key advances in fine-tuning techniques, dataset/domain handling, loss functions, and emerging alignment paradigms. We also highlight special-domain efforts (infants, limb-deficient persons, etc.) and their innovations in model design and training.

## Fine-Tuning Strategies in Foundation Models for Pose

Large pre-trained vision backbones (often ViT-based) are **fine-tuned** for pose by attaching lightweight task-specific heads and updating all or part of the model parameters. For example, Meta’ s **Sapiens** models (up to 2 billion parameters) use a ViT encoder pre-trained on 300M human images (self-supervised via MAE) and a small decoder head for pose; both encoder and decoder are then fine-tuned end-to-end on keypoint data [1] [2]. In practice, **differential learning rates** are used – lower LR on early (pre-trained) layers and higher LR on later layers – to preserve learned features while adapting to the pose task [3]. Sapiens reports that this simple pretrain-then-finetune approach yields excellent generalization, even when labeled data is scarce or synthetic [4] [5]. Layers are not frozen in Sapiens, but the gentle LR on the backbone acts similar to partial freezing.

Another example, **ViTPose++ (TPAMI 2023)**, demonstrates that a plain ViT encoder with a minimal decoder can be scaled from 20M to 1B parameters and fine-tuned to outperform prior CNN-based pose nets [6] [7]. ViTPose++ is extremely **flexible** in training paradigms – supporting top-down (with person crop) or bottom-up (direct multi-person) pose detection in one model [6] [8]. Importantly, ViTPose++ introduces a Mixture-of-Experts-style fine-tuning: it **factorizes knowledge** in the transformer by having **task-agnostic vs. task-specific Feed-Forward Networks (FFNs)** within each ViT block [9]. In other words, some FFN submodules are shared across tasks while others are specialized, enabling a single model to handle heterogeneous keypoint sets (e.g. human body vs. animal) without conflict [10] [11]. This adapter-like strategy lets ViTPose++ **fine-tune one foundation model on multiple pose tasks** simultaneously. Indeed, a single ViTPose++ model achieves state-of-the-art accuracy on COCO (human pose), COCO-WholeBody (whole-body pose), and AP-10K/APT-36K (animal pose) **at once**, matching or beating task-specific models [12]. This is achieved without sacrificing speed, thanks to the efficient sharing of most parameters [10].

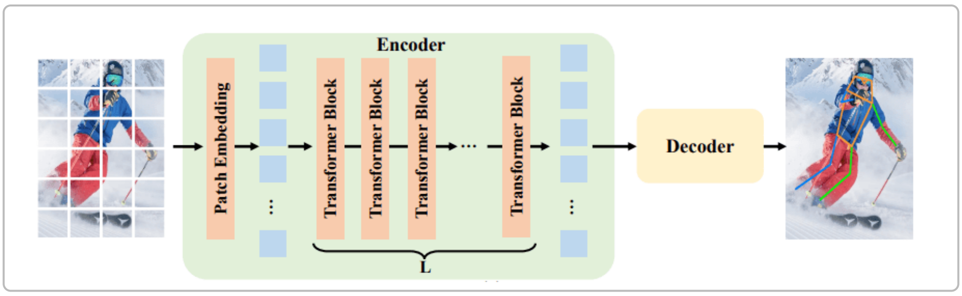

Figure 1: Vision Transformer-based pose estimation model (ViTPose architecture). A ViT encoder (with L transformer blocks) processes image patches, and a task-specific decoder outputs 2D keypoint heatmaps [8]. Large pre-trained ViT backbones (sometimes 1B+ params) can be fine-tuned for pose estimation and scaled to improve accuracy [9] [13].

Beyond architecture design, researchers have explored **parameter-efficient fine-tuning** for pose models, inspired by NLP. One CVPR 2024 work proposes "Sheared Backpropagation" to fine-tune large ViT models under memory constraints [14] [15]. It prunes and "freezes" a randomized subset of activation gradients during each backpropagation step (updating only the most important weights), dramatically reducing memory usage while maintaining accuracy [14] [16]. Such techniques allow fine-tuning a foundation pose model on modest GPUs ~20–60% faster with negligible accuracy drop [17] [15]. Although not specific to pose, this method makes fine-tuning billion-parameter pose models more feasible. In practice, though, recent pose works have more often fine-tuned fully or used simple adapters rather than LoRA modules – likely because large human datasets are available and full fine-tuning yields best results. For instance, Sapiens explicitly fine-tunes all encoder parameters (with a small LR decay per layer) rather than freezing any part [3].

Multi-stage fine-tuning has proven useful, especially for domain adaptation (discussed next). A clear example is the **two-stage distillation** in **DWPose** (ICCV 2023) for whole-body pose. First, a large teacher model (e.g. RTMPose-x) guides a smaller student (RTMPose-l) by distilling intermediate features and logits for all keypoints (including occluded ones) [18]. Notably, they do not ignore invisible keypoints during training – the teacher's predictions for occluded joints serve as soft targets, providing the student with reasonable "guesses" instead of no supervision [19]. They also apply a scheduled distillation weight decay to reduce reliance on the teacher over time [20]. Then, in a second stage, DWPose self-distills the student: they freeze the student's backbone and fine-tune only its head for a few more epochs, using the student's own prior outputs as targets (a "head-aware" self-KD) [21]. This quick second-stage fine-tune (just 20% of full training time) gave an extra boost in precision [21]. The overall effect is that DWPose's distilled small model surpassed its big teacher* (66.5 AP vs 65.3 AP on COCO-WholeBody) [22], setting a new state-of-art for whole-body pose with far fewer parameters. This underscores how multi-stage fine-tuning (teacher-guided then self-refining) can yield highly efficient yet accurate pose models [23] [22].

In summary, fine-tuning large pose models today often means **end-to-end training with minor modifications**: leveraging massive pre-training, adding minimal heads, and possibly mixing shared vs. specific units for multi-tasking. Heavy use of adapter modules or LoRA on pose models has not been widely reported in 2024 literature – likely because full fine-tuning is achievable and tends to maximize accuracy when ample data is available. However, efficient fine-tuners (like Sheared Backprop) and smart adapter designs (as in ViTPose++) are emerging to make foundation pose models more adaptable and scalable.

# Dataset Usage: Cross-Domain Adaptation and Generalization

Large pose models are typically fine-tuned on aggregated datasets to improve robustness and domain coverage. ViTPose++ for example was fine-tuned on a union of **multiple human pose datasets** (MS-COCO, MPII, AI Challenger, etc.) and even animal pose data, thanks to its ability to handle differing keypoint schemas [12]. This multi-dataset training is a form of **multi-domain adaptation**, enabling a single model to generalize across "domains" like COCO (mostly street photos), MPII (mostly sports), and animal images. Experiments showed that jointly training on diverse keypoint datasets let ViTPose++ maintain top accuracy on each, whereas separate models would be needed otherwise [12]. Similarly, the **Sapiens** foundation model emphasizes generalization: it was pre-trained on an extremely diverse set of in-the-wild human images (Humans-300M) and fine-tuned on a new comprehensive keypoint set (308 keypoints combining body, face, hands, feet) [24] [25]. As a result, Sapiens models perform robustly on varied input types (face close-ups, upper-body selfies, multi-person scenes, etc.) without needing domain-specific tweaks [13] [5]. In fact, Sapiens-2B achieves **state-of-the-art 2D pose accuracy** on the Humans-5K benchmark, outperforming prior specialized models by a large margin (+7.1 AP over DWPose-L) [26] [27]. The authors note it generalizes across **varying ages, multiple people, and challenging real-world backgrounds** due to the broad pre-training [28]. This indicates that sheer dataset scale and diversity during pre-training/fine-tuning can give strong cross-domain pose performance.

When true domain shifts occur (e.g. different imaging conditions or subject populations), specialized fine-tuning approaches have been proposed. One example is adapting pose models to **low-light images** (e.g. night-time or dark indoor scenes). An ECCV 2024 paper introduced a dual-teacher unsupervised adaptation for extremely low-light 2D pose [29] [30]. In **stage 1**, they fine-tune a pose model on normal (well-lit) images with heavy low-light augmentations (simulating darkness). In **stage 2**, they use unlabeled low-light photos and two complementary teacher networks to generate pseudo-labels: a "main" teacher (center-based pose detector) covers confidently visible people, while a second teacher focuses on any missed people (often faint/dark) [30]. By training the student on both sets of pseudo-labels – along with person-specific augmentation to challenge it – the model achieved a **+2.4 AP improvement** (≈+6.8% relative) on real low-light pose data, despite using zero labeled dark images [31]. This work exemplifies **cross-domain fine-tuning without ground-truth**: first leverage source data with simulated target conditions, then self-train on target domain with teacher guidance. It eliminates the need for expensive low-light keypoint annotation, yet closes much of the performance gap to a fully supervised model [29] [32].

Another notable domain adaptation is for **infant pose estimation**. Infants have very different body proportions and frequent self-occlusions compared to adults, making off-the-shelf pose models fail (trained on adult data). Bose et al. (CVPR 2025 W. ABAW) present **SHIFT**, an unsupervised domain adaptation pipeline to adapt a pre-trained adult pose model to infants [33] [34]. They start with a synthetic adult pose dataset model and then **fine-tune on unlabeled infant videos** using a **Mean Teacher** framework: a teacher model (EMA of the student) generates pseudo-label keypoints for infants, and consistency loss is enforced between teacher and student predictions under various augmentations [33] [35]. To handle the large anatomical gap, SHIFT introduces an **infant pose prior** (learned manifold of plausible infant poses) to penalize implausible keypoint configurations [36] [37]. They also add a **visibility consistency** module: aligning predicted keypoints with the infant's segmentation mask to ensure the pose fits the visible limbs [38] [39]. These additional regularizers serve as "baby domain knowledge" during fine-tuning. The result was a significant accuracy gain on infant pose benchmarks – far outperforming naive fine-tuning or generic domain adaptation methods [40] [41]. In short, for **cross-age adaptation**, combining self-training with pose priors and occlusion-awareness produced a robust infant pose estimator from an adult model.

Domain generalization (training a model to handle unseen domains without further adaptation) is also studied. One approach is **synthetic-to-real generalization**. For example, some works use synthetic human datasets to fine-tune a model and apply it to real-world images without labels, using techniques like keypoint-level augmentation and consistency training (as in SHIFT above, or earlier methods like UniFrame). The large foundation models themselves (like Sapiens) implicitly tackle domain generalization by training on extremely varied "real + synthetic" data – Sapiens used both in-the-wild photos and high-fidelity synthetic human renderings for depth, normals, etc., to ensure its pose network isn't biased to a narrow distribution [42] [43].

Finally, there is increasing attention to **inclusive pose estimation** for people with disabilities or unique attributes. ICCV 2023 introduced **LDPose**, the first pose dataset for **limb-deficient individuals** (persons missing an arm/leg or with amputation) [44]. Traditional pose models struggled here because standard keypoint formats assume all limbs present. LDPose redefines the task: models must predict **standard keypoints for present limbs and the endpoints of missing limbs** (essentially where an amputated limb terminates) [44]. The authors fine-tuned state-of-art pose estimators on this new data and proposed a specialized **Limb-Deficient Loss (LDLoss)** [45]. LDLoss explicitly helps the model distinguish between residual limb keypoints vs. normal ones by **contrastive learning** – it contrasts the error for a "stump" endpoint vs. an intact joint, to encourage the network to not confuse the two [46]. They also defined new evaluation metrics (LD Metrics) to fairly measure performance on residual-limb keypoints separately [47]. With these adaptations, the fine-tuned models could localize prosthetic or residual limb endpoints much more reliably. This is a prime example of dataset-driven fine-tuning: by collecting ~28k images of limb-deficient persons and modifying the loss/metrics, they created a pose estimator that **works for amputees** – a previously overlooked use-case [48] [49]. We anticipate future work extending foundation models to other domains: e.g. pose estimation for **elderly people** (who may have different posture or use assistive devices), or for **sports/athletes** where domain-specific motion patterns occur. The tools seen – multi-domain training, teacher-student adaptation, and custom losses – will be key to those applications.

## Loss Functions: Keypoint Localization and Visibility

The choice of loss function is crucial when fine-tuning large models on 2D keypoint estimation. Most approaches still rely on **heatmap-based losses**, where the model outputs a probability heatmap for each keypoint. A classic choice is Mean Squared Error (MSE) between predicted and ground-truth heatmaps (the ground truth is a Gaussian blob centered at the true keypoint). For instance, Sapiens' pose decoder produces $K$ heatmaps for $K$ keypoints and is fine-tuned to minimize **MSE loss** between predicted heatmaps and target heatmaps [50] [51]. This proved effective for its 308-keypoint whole-body predictions. The LearnOpenCV report on Sapiens confirms they use heatmaps and note: "Sapiens pose estimation model uses heatmaps for each keypoint… Each value in the heatmap represents the probability of that keypoint at that pixel", and it optimizes an MSE-based loss $L_{\text{pose}} = \mathrm{MSE}(Y, \hat{Y})$ [52] [53].

Some recent models have explored **alternative localization losses** to overcome limitations of MSE on heatmaps (such as quantization error or difficulty modeling multi-modal uncertainties). One innovation, used in the lightweight **RTMPose** series and adopted by DWPose, is **SimCC (Simulated Continuous Convolution)**. In SimCC, rather than producing a 2D heatmap, the model outputs two 1D probability distributions per keypoint (one for the x-coordinate, one for y). Each is a classification over discrete pixel locations [54] [21]. Essentially, SimCC treats keypoint localization as two classification tasks, allowing use of cross-entropy loss which can be easier to optimize than pixel-wise MSE. RTMPose reported that SimCC yields accuracy on par with high-res heatmaps but with lower computation. DWPose's distillation framework explicitly mentions that "RTMPose predicts keypoints with a SimCC algorithm that treats keypoint localization as classification for horizontal and vertical coordinates" [21]. They retain this scheme during distillation (the student's head outputs SimCC logits, and the teacher's soft logits supervise via KL/divergence loss). This indicates that **Cross-Entropy on coordinate logits** (SimCC) is becoming a popular loss for transformer-based pose models, especially when trying to reduce output resolution or when a differentiable argmax is needed.

Another line of work proposes learning the **distribution of keypoint offset errors**. A recent method called RLE (Regression via Learned Distribution, CVPR 2022) replaced the fixed Gaussian heatmap with a **learned latent distribution** produced via normalizing flows [55]. This essentially trains the model to output a custom heatmap shaped by data, improving direct coordinate regression. While not a 2024 method, it's noted in LocLLM's related work that RLE improved regression by learning a better loss distribution [55]. Such approaches could be combined with foundation models to refine their coordinate outputs, though current large models still mostly use standard heatmap or SimCC losses.

**Visibility and occlusion handling** is another important aspect. Datasets like COCO keypoints label each keypoint as 0=not present, 1=present but occluded, 2=visible. Many traditional pipelines simply ignore occluded keypoints in the loss (i.e. do not penalize the model for not detecting a fully hidden joint). However, ignoring them entirely means the model never learns to predict reasonable positions for hidden joints. Some methods introduce an auxiliary **visibility classification loss** – e.g. having the model output a visibility flag per keypoint with a binary cross-entropy. But this is relatively rare in recent large models; more common is weighting the keypoint loss by visibility (weight = 0 for missing joints, maybe lower weight for occluded). For example, earlier works like OpenPose had a visibility output, but ViTPose/Sapiens do not explicitly mention one, implying they likely train only on labeled (visible) joints.

Interestingly, **DWPose's distillation** addresses visibility in a new way: as mentioned, it uses teacher predictions for invisible keypoints as training targets [56]. This effectively provides supervision on occluded joints where ground truth was absent (since the teacher, being a strong model, can infer a likely position). This strategy improved performance on face and hands, where occlusions are frequent [54] [21]. So rather than a separate loss, they integrate invisible keypoints into the main loss via teacher guidance. This is a form of **imputation** for occluded data.

Special domains provoke new loss terms. We saw **Limb-Deficient Loss (LDLoss)** in LDPose, which adds a term to better handle missing limbs. Specifically, LDLoss "contrasts residual vs intact limb keypoints," ensuring the model's feature for an amputated stump doesn't get confused with an actual wrist/ankle, for example [46]. Implementationally, this can be a contrastive loss or a re-weighting that penalizes mistakes on residual joints more. The effect is the model learns a clearer representation for "amputated" endpoints. This is a good reminder that as pose models are fine-tuned to new populations, the loss functions may need adjustment to encode the right priors (e.g. "missing limb endpoint is a different class of keypoint").

For **multi-person pose estimation**, additional losses come into play in bottom-up models: namely **association losses**. Bottom-up approaches (like HrHRNet or some ViT bottom-up variants) predict all keypoints and an **embedding tag** per keypoint to group them into individuals. The training uses a tag loss (often pulling same-person joints together and pushing different-person joints apart). ViTPose is primarily top-down, but it notes that for bottom-up mode they also predict **associate embedding maps** and train with the corresponding grouping loss [8] [57]. Large models can handle bottom-up, but the fine-tuning must include these multi-person grouping losses to be effective. In practice, top-down remains dominant in recent foundation models (as they often rely on an off-the-shelf detector for persons), which simplifies the loss to per-keypoint terms only.

In summary, **keypoint heatmap MSE** (or cross-entropy) remains the workhorse loss for fine-tuning, sometimes augmented with **coordinate classification** (SimCC) or **auxiliary losses** for special cases (visibility, grouping, etc.). The fine-tuning of foundation models hasn't required radically new loss functions, but rather careful application of known ones with perhaps **task-specific tweaks**. As one CVPR 2024 tutorial put it, "the decoder is trained with the same keypoint heatmap losses as traditional models, thanks to the minimal changes in architecture". The success of large models is less about novel losses and more about stronger features and more data – but as we broaden pose estimation's scope, loss functions are adjusted to guide these powerful models (e.g. LDLoss for amputees or pose priors for infants ensure the large model doesn't learn erroneous cues).

## Post-Training Alignment and Emerging Paradigms

While 2D pose estimation is a straightforward supervised task (with clear objectives like minimizing keypoint error), researchers have started to explore **"alignment" techniques inspired by large language models** – e.g. instruction tuning, multi-modal reasoning, and even RL-based refinement – to push pose models further. A prime example is the CVPR 2024 work **LocLLM** by Wang et al. [58] [59]. **LocLLM** is a Large Language Model (LLM)-powered pose estimator that treats keypoint detection as an **instruction-following problem**. It integrates a vision encoder with a pre-trained LLM (similar to GPT) such that you can "ask" the model in natural language about keypoint locations. For instance, given an image, one can input: "Locate the person's left elbow" and the model will output the coordinates. During training, LocLLM was fine-tuned with **localization-based instruction conversations**: they generated text prompts describing each keypoint and trained the model to produce the correct coordinates as the "answer" [58] [60]. Essentially, the LLM's reasoning capability is harnessed to improve generalization – it learns **spatial relationships and keypoint definitions from text**. Importantly, LocLLM introduced a **parameter-efficient tuning** approach (e.g. using a Q-Former or LoRA adapters) so that the huge LLM (billions of params) didn't need full fine-tuning [60]. Only small portions were tuned to align the visual features with text tokens.

The results were striking: LocLLM achieved **77.4 AP on COCO** (close to state-of-art ConvNet/ViT models) and showed superior cross-dataset generalization [61] [59]. For example, when trained on COCO and evaluated on MPII (without fine-tune), it scored 33.4 PCK@0.1, beating ViTPose by +7.8 points [59]. It could also localize **novel keypoints via text** – e.g. the model was never trained on "pelvis" keypoint (COCO has no pelvis joint) but if you prompt "pelvis", it can output a reasonable location [62]. This zero-shot ability comes from the LLM's encoded human anatomy knowledge plus the learned instruction tuning. In essence, LocLLM is **instruction-tuned** on vision-language data so it can follow human commands about poses. This is a new paradigm for pose estimation: moving from a fixed set of keypoints to a flexible, user-driven description. It opens the door to interactive pose estimation (asking for certain keypoints or formats) and potentially using **RLHF (Reinforcement Learning from Human Feedback)** down the line to align pose models with user preferences (e.g. emphasize certain joints or maintain physically plausible output). So far, we have not seen RLHF applied to 2D pose – likely because "correct keypoint positions" are objective and don't require subjective human preference tuning. But related ideas like **rewarding physical plausibility** have appeared (e.g. RL to enforce no jitter in pose tracking, or to ensure contact points in 3D pose). In 2D, a recent concept is "PoseGPT" or "PoseChat" models that could take instructions like "find the person doing XYZ in the image and output their pose". Such models would need instruction fine-tuning similar to LocLLM and perhaps a form of reward model to ensure the outputs are useful to humans (for instance, aligning pose outputs with descriptive text or task success).

Another way alignment shows up is in **unifying pose with generative models**. For instance, **HumanSD (ICCV 2023)** fine-tuned a Stable Diffusion model to take a text prompt and a pose skeleton as input to generate human images [63]. While not pose estimation, it required aligning the diffusion model's image output with the 2D pose condition (using a pose encoder and a special conditioning loss). Techniques like such **pose-conditioned diffusion** might eventually incorporate RLHF (to judge image realism with the given pose) or instruction tuning (so a user can prompt "generate a person in this pose"). From the estimation side, one might imagine an **instruction-tuned pose model that can explain its predictions** ("the left knee is occluded behind the table") – which merges visual reasoning with pose output.

In summary, post-training alignment in the strict sense (RLHF and reward modeling) hasn't yet become mainstream for pose estimators, because the task is typically solved with supervised learning on keypoint labels. However, the influence of the LLM revolution is clearly reaching pose estimation:

LocLLM’s **multimodal instruction tuning** is a pioneering example [58] [59]. It demonstrates that giving a foundation model the ability to understand **human-language descriptions of keypoints** can dramatically improve generalization and flexibility (e.g. adding new keypoints on the fly) [64]. We expect more works to follow this direction, possibly integrating pose estimation into larger vision-language models or employing **chat-based refinement** (e.g. a human could give feedback like “the pose is wrong, the arm should be higher” and an RLHF-trained model might adjust the output). These are early days for such approaches, but LocLLM’s success suggests that **bridging pose models with language and “knowledge”** is a very promising avenue. In effect, it treats the pose model as an “agent” that can reason about positions (“the knee is below the hip, at this pixel coordinate”) rather than a pure feed-forward predictor. This could help tackle data scarcity (by using text to inject prior knowledge) and improve robustness in unusual scenarios via logical constraints.

## Conclusions and Outlook

Fine-tuning large vision models has enabled 2D pose estimation to reach new heights in accuracy and generality. Techniques like fully end-to-end training on broad data (Sapiens), multi-task knowledge sharing (ViTPose++), and teacher-student distillation (DWPose) address the technical challenges of adapting huge backbones to keypoint prediction. Meanwhile, tackling who and where these models work has led to domain-focused advances – from unsupervised adaptation for infants or dark images, to new datasets for limb differences – each with clever fine-tuning tricks to handle the distribution shifts. Loss functions have largely followed established best practices (heatmap regression, with refinements like SimCC or special occlusion treatments) and have been sufficient for these large models when coupled with enormous data.

Perhaps most intriguing is the foray of pose estimation into multimodal and interactive AI. The year 2024 saw the first LLM-aligned pose model that can respond to language instructions and leverage language priors [58] [59]. This hints that future pose estimators might be more **flexible intelligent systems** rather than single-purpose point predictors. Imagine a system that can both detect poses and answer questions about them (“who is bending their knees more?”) or correct them based on feedback – these are tasks that bridging with language models can facilitate. Though not heavily using RLHF yet, such systems may eventually incorporate human feedback loops to ensure the pose outputs are not just accurate, but also useful and aligned with human intent (especially in applications like assisted exercise, physical therapy, or content creation).

In closing, the trend in 2D pose estimation research is clear: leverage the power of **big models and big data**, but fine-tune and adapt them cleverly to each scenario. The works highlighted (ViTPose++, Sapiens, LocLLM, and others) represent representative milestones. Sapiens showed that a gigantic human-centric model can generalize across domains and tasks with simple fine-tuning [4] [1]. ViTPose++ demonstrated how to **fine-tune one transformer to master many keypoint types at once** [9] [11]. DWPose illustrated new training schedules that squeeze more out of models via distillation [65] [22]. And LocLLM has opened the door to instruction-aware pose models that break the mold of fixed outputs [58] [59]. Each of these expands the capabilities of pose estimation: more scalable, more general, more efficient, and more interactive.

**Original Papers:**


- **ViTPose & ViTPose++ (Generic Pose Transformers):** Y. Xu et al., “ViTPose: Simple Vision Transformer Baselines for Human Pose Estimation” (NeurIPS 2022) and “ViTPose++: Vision Transformer Foundation Model for Generic Body Pose Estimation” (TPAMI 2023) [10] [12].

- **Sapiens (Meta' s Human Vision Foundation):** R. Khirodkar et al., "Sapiens: A Foundation for Human Vision Models" (ECCV 2024) 4 1 .
- **DWPose (Whole-Body Distillation):** Z. Yang et al., "Effective Whole-Body Pose Estimation with Two-Stages Distillation" (ICCV 2023 workshop) 23 22 .
- **LDPose (Limb-Deficient Pose):** J. Ying et al., "LDPose: Towards Inclusive Human Pose Estimation for Limb-Deficient Individuals" (ICCV 2023) 48 47 .
- **SHIFT (Infant Pose UDA):** S. Bose et al., "Leveraging Synthetic Adult Datasets for Unsupervised Infant Pose Estimation (SHIFT)" (CVPR 2025 workshop) 33 37 .
- **Low-Light Pose:** Y. Ai et al., "Domain-Adaptive 2D Human Pose Estimation via Dual Teachers in Extremely Low Light" (ECCV 2024) 30 31 .
- **LocLLM (Pose via LLM):** Y. Wang et al., "LocLLM: Exploiting Generalizable Human Keypoint Localization via Large Language Model" (CVPR 2024) 58 59 .
- **Sheared Backprop (Efficient Tuning):** Z. Yu et al., "Sheared Backpropagation for Fine-tuning Foundation Models" (CVPR 2024) 14 16 . Each of these works offers insights into fine-tuning large models for pose – setting a roadmap for future research in **scalable, adaptable, and human-aligned pose estimation**.

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